

(1001CJA621121003)



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

Test Pattern

JEE(Advanced)

REVIEW TEST

14-03-2021

JEE(Main + Advanced) : NURTURE COURSE (PHASE : TEPS)

ANSWER KEY

PAPER-1

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	B	D	C	C	B	C,D	B,C	B,C	A,B,D
	Q.	11	12								
	A.	A,C,D	C,D								
SECTION-II	Q.	1	2	3	4	5	6				
	A.	0.30 to 0.31	6.00	0.20 to 0.21	9.00	113.33	1.80				

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	C	C	A	B	A	A,C,D	A,B	A,C	A,B,C
	Q.	11	12								
	A.	A,C,D	B,C,D								
SECTION-II	Q.	1	2	3	4	5	6				
	A.	3.00	6.00	2.40	6.00	0.45	-597.51				

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	B	B	A	B	D	B,C,D	B,C	A,B,C	A,B,C,D
	Q.	11	12								
	A.	A,B,C	B,C								
SECTION-II	Q.	1	2	3	4	5	6				
	A.	81.00	669.00	96.50	1.60	6.00	13.00				

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ANSWER KEY

PAPER-2

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6				
	A.	A,B,C,D	A,C	A,D	AB	AD	AC				
SECTION-II	Q.	1	2	3	4	5	6				
	A.	4.70 to 4.79	1751.00 to 1752.00	910.00	1.93 to 1.94	2.50 to 2.60	25.00				
SECTION-III	Q.	1	2	3	4	5	6				
	A.	5	8	4	3	8	6				

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6				
	A.	A,B,C,D	A,B,D	B,C,D	A,C,D	A,B,C,D	A,C				
SECTION-II	Q.	1	2	3	4	5	6				
	A.	640.00	4.50	2.75	1.25	5.00	8.00				
SECTION-III	Q.	1	2	3	4	5	6				
	A.	5	5	7	3	9	6				

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6				
	A.	B,C	B,C,D	A,B,C	B,C,D	B,C	A,B,C,D				
SECTION-II	Q.	1	2	3	4	5	6				
	A.	8.00	18.00	128.00	72.00	41.00	96.00				
SECTION-III	Q.	1	2	3	4	5	6				
	A.	4	7	4	9	4	2				



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PART-2 : CHEMISTRY

SOLUTION

SECTION-I

1. Ans.(C)
2. Ans.(C)
3. Ans. (C)
4. Ans. (A)
5. Ans.(B)
 $n\text{C}_2\text{H}_4(\text{g}) \rightarrow (\text{C}_2\text{H}_4)_n(\text{g}) ; \Delta H = -6000 \text{ kJ}$
 $-6000 = (n \times 540) - (2n \times 330) \Rightarrow n = 50$
6. Ans.(A)

Sol. Let moles of each compound $-x$

$$2x + 8x = 0.5V_1 \quad \dots(1)$$

$$7x = 0.2V_2 \quad \dots(2)$$

from (1) and (2);

$$\left(\frac{10}{7}\right) = \frac{5V_1}{2V_2} \Rightarrow \frac{V_1}{V_2} = \left(\frac{4}{7}\right)$$

7. Ans.(A,C,D)

When $x = 1$, liquid is pure 'A' and P

$$= P_A^0 = 100 \text{ cm Hg}$$

When $x = 0$, liquid is pure 'B' and P

$$= P_B^0 = 60 \text{ cm Hg}$$

$$\text{Now, } y = \frac{P_A}{P_{\text{total}}} = \frac{x \times P_A^0}{P_{\text{total}}}$$

$$= \frac{0.5 \times 100}{(60 + 40 \times 0.5)} = 0.625$$

$$\text{and, } \frac{1}{P_{\text{total}}} = \frac{y}{P_A^0} + \frac{1-y}{P_B^0} = \frac{0.5}{100} + \frac{1-0.5}{60}$$

$$\therefore P_{\text{total}} = P = 75 \text{ cm Hg}$$

8. Ans. (A, B)

On increasing the volume, reaction will

shift in forward direction hence moles of $A(\text{s})$ will decrease and moles of $B(\text{g})$ will increase. But concentration will not change because rate of forward reaction is not changing.

9. Ans.(A,C)
10. Ans.(A,B,C)
11. Ans.(A,C,D)
12. Ans.(B,C,D)

SECTION-II

1. Ans.(3.00)
2. Ans.(6.00)
3. Ans.(2.40)
4. Ans.(6.00)
5. Ans.(0.45)

Sol. For minimum pressure, the process should be irreversible. Now,

$$\Delta U = w$$

$$\text{or, } n \cdot C_{v,m} \cdot (T_2 - T_1) = -P_{\text{ext}} (V_2 - V_1)$$

$$\text{or, } n \frac{5}{2} R \cdot (600 - 300)$$

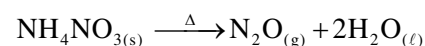
$$= -P_2 \left(\frac{nR \cdot 600}{P_2} - \frac{nR \cdot 300}{P_1} \right)$$

$$\Rightarrow P_2 = 0.45 \text{ MPa}$$

6. Ans.(-597.51)

$$\text{Sol. } \Delta u \times \left[\frac{1}{80} \right] = (1.25 \times 6) \text{ kJ}$$

$$\Rightarrow \Delta u = (80 \times 1.25 \times 6) \text{ kJ} = 600 \text{ KJ}$$



$$\Delta n_g = 1$$

$$\Delta H = \Delta u + (\Delta n_g)RT$$

$$= -600 \text{ kJ} + \left(\frac{1 \times 8.314 \times 300}{1000} \right) \text{ kJ}$$

$$= (-600 + 2.4942) \text{ kJ} = -597.5 \text{ kJ}$$

PART-3 : MATHEMATICS

SOLUTION

SECTION-I

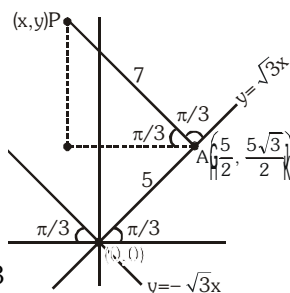
1. **Ans. (D)**

Sol. From the figure

$$x = \frac{5}{2} - 7 \cos \frac{\pi}{3} = -1$$

$$y = \frac{5\sqrt{3}}{2} + 7 \sin \frac{\pi}{3} = 6\sqrt{3}$$

$$\therefore PQ = \sqrt{121 + 48} = 13$$



2. **Ans. (B)**

$$\text{Sol. } \left(2^{\log_2(x^2-3x+3)}\right)^{\log_2(x^2-3x+3)} + (x^2-3x+3)^{\log_2(x^2-3x+3)} = 2$$

$$\Rightarrow 2(x^2-3x+3)^{\log_2(x^2-3x+3)} = 2$$

$$\Rightarrow (x^2-3x+3)^{\log_2(x^2-3x+3)} = 1$$

Taking log both side to the base 2

$$\Rightarrow (\log_2(x^2-3x+3))^2 = 0$$

$$\Rightarrow x^2-3x+3 = 1$$

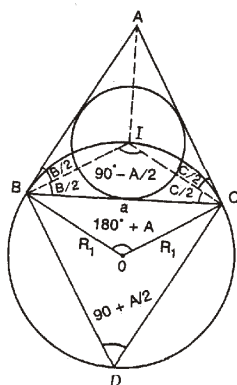
$$\Rightarrow x^2-3x+2 = 0$$

$$\Rightarrow x = 2, 1$$

3. **Ans. (B)**

$$\angle BIC + \angle BDC = 180^\circ$$

$$\therefore \angle BDC = 90^\circ + \frac{A}{2}$$



Let O be the centre of the required circle and R_1 is the radius and I be the incentre of ΔABC then

$$\angle BDC = 2 \angle BIC = (180^\circ + A)$$

and side $BC = a$

In ΔBOC ,

$$\cos(180 + A) = \frac{R_1^2 + R_1^2 - a^2}{2R_1 R_1}$$

$$\Rightarrow -2 R_1^2 \cos A = 2 R_1^2 - a^2$$

$$\Rightarrow a^2 = 2 R_1^2 (1 + \cos A)$$

$$\Rightarrow a^2 = 4 R_1^2 \cos^2 \frac{A}{2}$$

$$\Rightarrow R_1^2 = \frac{a^2}{4} \sec^2 \frac{A}{2}$$

$$\Rightarrow R_1 = \frac{a}{2} \sec \frac{A}{2}$$

4. **Ans. (A)**

$$a_0 = \cos \theta, a_1 = \sqrt{\frac{1}{2}(1 + \cos \theta)} = \cos \frac{\theta}{2}$$

$$a_2 = \sqrt{\frac{1}{2}\left(1 + \cos \frac{\theta}{2}\right)} = \cos \frac{\theta}{2^2}$$

.....

$$a_n = \frac{\cos \theta}{2^n}$$

$$\Rightarrow \frac{\sqrt{1-a_0^2}}{(a_1 a_2 \dots a_n) \times \sqrt{1-a_n^2}}$$

$$= \frac{\sqrt{1-\cos^2 \theta}}{\cos \frac{\theta}{2} \cos \frac{\theta}{2^2} \dots \cos \frac{\theta}{2^n} \sqrt{1-\cos^2 \frac{\theta}{2^n}}}$$

$$= \frac{\sin \theta}{\left(\frac{\sin \left(2^n \cdot \frac{\theta}{2^n}\right)}{2^n \sin \frac{\theta}{2^n}}\right) \sin \left(\frac{\theta}{2^n}\right)} = 2^n$$

5. **Ans. (B)**

$$\text{On solving, } y = \frac{x^3}{(x-\alpha)(x-\beta)(x-\gamma)}$$

$$\text{Also } x^3 - 3x^2 + 2x + 4 = (x-\alpha)(x-\beta)(x-\gamma).$$

At $x = 2$,

$$y = \frac{2^3}{(2-\alpha)(2-\beta)(2-\gamma)} = \frac{8}{2^3 - 3 \cdot 2^2 + 2 \cdot 2 + 4} = 2$$

6. **Ans. (D)**

As sum of two sides of a triangle is greater than the third side

$$1 + \frac{b-c}{a}, 1 + \frac{c-a}{b} \text{ \& } 1 + \frac{a-b}{c}$$

are all positive

Using $AM \geq GM$, we get

$$\frac{a\left(1 + \frac{b-c}{a}\right) + b\left(1 + \frac{c-a}{b}\right) + c\left(1 + \frac{a-b}{c}\right)}{a+b+c}$$

$$\geq \left[\left(1 + \frac{b-c}{a} \right)^a \left(1 + \frac{c-a}{b} \right)^b \left(1 + \frac{a-b}{c} \right)^c \right]^{\frac{1}{a+b+c}}$$

$$\Rightarrow 1 \geq \left[\left(1 + \frac{b-c}{a} \right)^a \left(1 + \frac{c-a}{b} \right)^b \left(1 + \frac{a-b}{c} \right)^c \right]^{\frac{1}{a+b+c}}$$

$$\Rightarrow \left(1 + \frac{b-c}{a} \right)^a \left(1 + \frac{c-a}{b} \right)^b \left(1 + \frac{a-b}{c} \right)^c \leq 1$$

But given

$$\left(1 + \frac{b-c}{a} \right)^a \left(1 + \frac{c-a}{b} \right)^b \left(1 + \frac{a-b}{c} \right)^c \geq 1$$

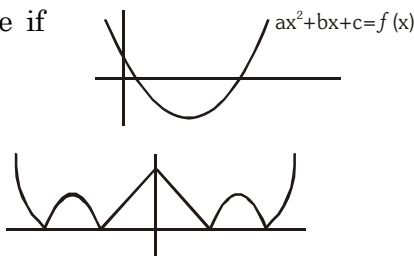
which is possible only if

$$\left(1 + \frac{b-c}{a} \right)^a \left(1 + \frac{c-a}{b} \right)^b \left(1 + \frac{a-b}{c} \right)^c = 1$$

i.e. $b-c=c-a=a-b=0$ or $a=b=c$

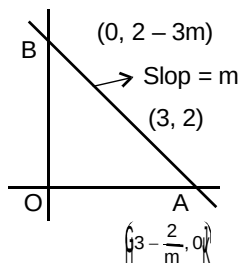
7. **Ans. (B,C,D)**

Possible if



8. **Ans. (B,C)**

$$S_{\min} = 2.3.2 = 12$$



$$OA + OB = 5 - \left(3m + \frac{2}{m} \right)$$

$$= 5 + \frac{\sqrt{2}}{m} - \sqrt{-3m} + 2\sqrt{6}$$

$$\text{Hence } (OA + OB)_{\min} = 5 + 2\sqrt{6}$$

Hence (B) (C)

9. **Ans. (A,B,C)**

$$2(\sin x - \cos 2x) - \sin 2x (1 + 2\sin x) + 2\cos x = 0$$

$$\Rightarrow 2\sin x - \sin 2x - 2\cos 2x - 2\sin x \sin 2x + 2\cos x = 0$$

$$\Rightarrow 2\sin x - \sin 2x - 2\cos 2x - (\cos x - \cos 3x) + 2\cos x = 0$$

$$\Rightarrow 2\sin x (1 - \cos x) + 4\cos^3 x - 3\cos x + \cos x - 2(2\cos^2 x - 1) = 0$$

$$\Rightarrow 2\sin x(1 - \cos x) + 4\cos^3 x - 4\cos^2 x - 2\cos x + 2 = 0$$

$$\Rightarrow 2\sin x(1 - \cos x) - 4\cos^2 x(1 - \cos x) + 2(1 - \cos x) = 0$$

$$\Rightarrow (1 - \cos x) \{2\sin x - 4(1 - \sin^2 x) + 2\} = 0$$

$$\Rightarrow \cos x = 1 \text{ or } \sin x - 2(1 - \sin^2 x) + 1 = 0$$

$$\Rightarrow x = 2n\pi \quad (2\sin x - 1)(\sin x + 1) = 0$$

$$\sin x = \frac{1}{2} \text{ or } \sin x = -1$$

$$x = 2n\pi, x = n\pi + (-1)^n \frac{\pi}{6} \text{ or } x = 2n\pi - \frac{\pi}{2}$$

10. **Ans. (A,B,C,D)**

Consider $\tan 20^\circ - 2\tan 10^\circ$

$$= \frac{\sin 20^\circ \cos 10^\circ - 2 \sin 10^\circ \cos 20^\circ}{\cos 10^\circ \cos 20^\circ}$$

$$= \frac{(\sin 30^\circ + \sin 10^\circ) - 2(\sin 30^\circ - \sin 10^\circ)}{2 \cos 10^\circ \cos 20^\circ}$$

$$= \frac{3\sin 10^\circ - \sin 30^\circ}{2 \cos 10^\circ \cos 20^\circ} = \frac{4\sin^3 10^\circ}{2 \cos 10^\circ \cos 20^\circ} > 0$$

$$\Rightarrow \tan 20^\circ > 2\tan 10^\circ$$

$$\therefore y > x \text{ \& } z > w \quad \dots (i)$$

Now $2w - 2y$

$$= 2\tan 10^\circ + \tan 70^\circ - \tan 20^\circ - \tan 50^\circ$$

$$= 2\tan 10^\circ + \tan 20^\circ \tan 50^\circ \tan 70^\circ$$

$$= 2\tan 10^\circ + \tan 50^\circ$$

$$= 2x \text{ which is positive}$$

$$\therefore w = x + y \text{ \& } w > y \quad \dots (ii)$$

Combining (i) & (ii) $z > w > y > x$

Also $2y - z$

$$= \tan 20^\circ + \tan 50^\circ - \frac{1}{2} \tan 20^\circ - \frac{1}{2} \tan 70^\circ$$

$$= \frac{1}{2} \tan 20^\circ + \tan 50^\circ - \frac{1}{2} \tan 70^\circ$$

$$= \frac{1}{2} (\tan 20^\circ + 2 \tan 50^\circ - \tan 70^\circ)$$

$$= \frac{1}{2} \left(\frac{\sin 70^\circ}{\cos 20^\circ \cos 50^\circ} - \frac{\sin 20^\circ}{\cos 50^\circ \cos 70^\circ} \right)$$

$$= \frac{\sin 140^\circ - \sin 40^\circ}{4 \cos 20^\circ \cos 50^\circ \cos 70^\circ} = 0$$

11. **Ans. (A,B,C)**

$$c_2 \rightarrow c_2 - c_1 x$$

$$\begin{vmatrix} a_1 + b_1 x & b_1(1-x^2) & c_1 \\ a_2 + b_2 x & b_2(1-x^2) & c_2 \\ a_3 + b_3 x & b_3(1-x^2) & c_3 \end{vmatrix} = (1-x^2) \begin{vmatrix} a_1 + b_1 x & b_1 & c_1 \\ a_2 + b_2 x & b_2 & c_2 \\ a_3 + b_3 x & b_3 & c_3 \end{vmatrix}$$

$$c_1 \rightarrow c_1 - c_2 x = (1-x^2) \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0$$

12. Ans. (B,C)

given lines $2x - y + 4 = 0$

and $-x + 2y + 1 = 0$

angle bisector $\frac{2x-y+4}{\sqrt{5}} = \pm \frac{(-x+2y+1)}{\sqrt{5}}$

$a_1a_2 + b_1b_2 < 0$

$\therefore$ for acute angle bisector take +ve sign

$\therefore$ acute angle bisector $x - y + 1 = 0$

for obtuse angle bisector take -ve sign

$\therefore$ obtuse angle bisector $x + y + 5 = 0$

for origin take +ve sign $\Rightarrow$ acute angle of the lines acute angle between

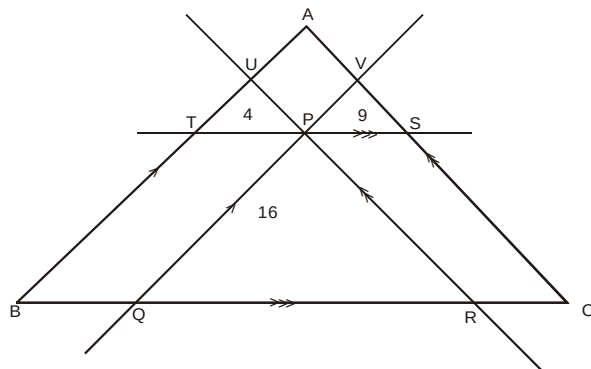
$$L_1 \& L_2 = \tan^{-1} \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$= \tan^{-1} \left| \frac{2 - \frac{1}{2}}{1 + 2 \cdot \frac{1}{2}} \right| = \tan^{-1} \left(\frac{3}{4} \right)$$

SECTION-II

1. Ans. 81.00

ΔPQR , ΔPTU & ΔPSV are similar to ΔABC
& Let area of ΔABC be A



$$\text{Hence } \frac{QR}{BC} = \sqrt{\frac{16}{A}}, \quad \frac{PT}{BC} = \sqrt{\frac{4}{A}}$$

$$\& \frac{SP}{BC} = \sqrt{\frac{9}{A}}$$

$$\text{Hence } \frac{QR + PT + SP}{BC} = \frac{4 + 2 + 3}{\sqrt{A}} = 1$$

(as $PT = BQ$ & $SP = CR$)

$\Rightarrow A = 81$ sq. units

2. Ans. 669.00

Sol. 1, 11, 111, 1111, 11111, 111111, only 3rd, 6th,

$$9^{\text{th}} \text{ number divisible} = \frac{2008}{3} = 669.33$$

3. Ans. 96.50

Let $f(x) = ax^2 + bx + c$

$a = 1$, $c = f(0) = 48$

$$f(x) = x^2 + bx + 48$$

$$f(3) = f(-3) \Rightarrow b = 0$$

$$f(x) = x^2 + 48$$

$$f(x+1) = (x+1)^2 + 48$$

$$\therefore g(x) = (f(x) + f(x+1)) = 2x^2 + 2x + 97$$

$$\text{minimum value} = -\left(\frac{4 - 4.2.97}{4.2}\right) = \frac{193}{2}$$

4. Ans. 1.60

Sol. Do yourself

5. Ans. 6.00

$$t = \left(\frac{1}{1-a}\right) \left(\frac{1}{1-b}\right) \left(\frac{1}{1-c}\right)$$

where a, b, c are roots of $12x^3 - 4x^2 - 3x + 1 = 0$

$$\Rightarrow 12 \left(x^3 - \frac{x^2}{3} - \frac{x}{4} + \frac{1}{12} \right) = 12(x-a)(x-b)(x-c)$$

$$\Rightarrow (1-a)(1-b)(1-c) = \frac{1}{2}$$

$$\Rightarrow t = 2$$

Now $y = \log_{p+2}(3x^2 + 2x + 6 - p)$ is defined
 $\forall x \in \mathbb{R}$

$$\Rightarrow p + 2 > 0, p + 2 \neq 1 \& 3x^2 + 2x + 6 - p > 0$$

$$\Rightarrow p > -2, p \neq -1 \& 4 - 12(6 - p) < 0$$

$$\text{or } -2 < p < \frac{17}{3}, p \neq -1$$

$\therefore$ Number of integral values of p is 7.

6. Ans. 13.00

$$a + b + c = 6 \cdot \frac{\sin A + \sin B + \sin C}{3}$$

$$\Rightarrow 2R(\sin A + \sin B + \sin C)$$

$$= 2(\sin A + \sin B + \sin C)$$

$$\Rightarrow R = 1$$

Now, $a = 2R \sin A$

$$1 = 2 \times 1 \sin A$$

$$\{\cos(\cot^{-1}(\cos A)) > 0\}$$

$$\Rightarrow \angle A = 30^\circ$$

$$\Rightarrow \cos A > 0$$

$$b = 2R \sin B \quad \therefore A < 90^\circ\}$$

$$\frac{\sqrt{3}-1}{\sqrt{2}} = 2 \times 1 \sin B \Rightarrow \sin B = \frac{\sqrt{3}-1}{2\sqrt{2}}$$

$$\angle B = 15^\circ$$

$$\therefore \angle C = 135^\circ$$

Now,

$$\Sigma \sin^2 2A + 2 \Pi \sin 2A \sin 2B$$

$$= (\sin 2A + \sin 2B + \sin 2C)^2$$

$$= \left(\frac{\sqrt{3}}{2} + \frac{1}{2} - 1 \right)^2 = \left(\frac{\sqrt{3}-1}{2} \right)^2 = 1 - \frac{\sqrt{3}}{2}$$

$$= 1 - 0.866 = 0.134$$

$$\therefore 100 \times 0.134$$

$$\therefore [13.4] = 13$$



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PART-2 : CHEMISTRY

SOLUTION

SECTION-I

1. Ans.(A,B,C,D)
2. Ans.(A,B,D)
3. Ans.(B,C,D)
4. Ans.(A,C,D)
5. Ans.(A,B,C,D)

$$\log \frac{k_{1000K}}{k_{500K}} = \frac{9200}{2.3 \times 2} \left[\frac{1}{500} - \frac{1}{1000} \right]$$

$$\Rightarrow k_{1000K} = 6.93 \times 10^{-2} \text{ s}^{-1}$$

$$\Rightarrow t_{1/2} = 10 \text{ sec.}$$

6. Ans.(A,C)

Sol. The distance between nearest neighbours is same in both forms means '2r' is same in both and hence the unit cell edge length, i.e., cell constant, a, is different in both.

$$a_{\alpha} = \frac{4r}{\sqrt{3}} \text{ and } a_{\beta} = \frac{4r}{\sqrt{2}}$$

The distance between next nearest neighbours in both form is 'a', which is different.

Void space in BCC = 32%, FCC = 26%

In both forms, the required diameter of largest spheres = (a - 2r) and 'a' is different

SECTION-II

1. Ans.(640.00)

Sol. $T = m \times V + C$

$$200 = m \times 20 + C$$

$$800 = m \times 40 + C$$

$$600 = 20 m \Rightarrow m = \frac{600}{20} = 30; C = 200 - 20 m$$

$$C = 200 - 20 \times 30 = 200 - 600 = -400$$

$$T = 30V - 400$$

$$\frac{PV}{R} = 30 V - 400$$

$$\text{or, } P = 30 R - \frac{400R}{V}$$

$$W = \int_{20}^{40} -PdV = \int_{20}^{40} \left(\frac{400R}{V} - 30R \right) dV$$

$$= 400 R [\ln V]_{20}^{40} - 30 R (40 - 20)$$

$$= 400 R \times \ln 2 - 600 R$$

$$= 280 R - 600 R = -320 R \Rightarrow -640 \text{ Cal}$$

2. Ans.(4.50)

3. Ans.(2.75)

4. Ans.(1.25)

5. Ans.(5.00)

6. Ans.(8.00)

SECTION-III

1. Ans.(5)

2. Ans.(5)

3. Ans.(7)

4. Ans.(3)

5. Ans.(9)

Sol. ${}_Z A^M \rightarrow (Z-6) B^{(M-12)} + 3_2 \alpha^4$
 $t = 0 \quad 1 \text{ mole} \quad 3 \times 0.75 \text{ (mole)}$

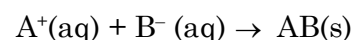
$t = 20 \text{ days} \quad 0.25 \text{ mole}$

Volume of He

$$= (3 \times 0.75 \times 22.4) \ell = (5.6 \times x) \ell$$

$$\Rightarrow x = 9$$

6. Ans.(6)



$$\Delta_f H^0 (\text{kcal/mole}) \quad 4x \quad -5x \quad -21.5$$

$$\Delta_f H^0 = -20 = (-21.5) - (4x - 5x)$$

$$x = 1.5$$

$$(\Delta_f H^0)_{A^+} = 4x = 6$$

PART-3 : MATHEMATICS

SOLUTION

SECTION-I

1. Ans. (B,C)

Sol. $\tan\left(B - \frac{\pi}{2}\right) = -3$

$$\Rightarrow \tan B = \frac{1}{3}$$

$$\tan\left(\frac{3\pi}{2} - A\right) = \frac{-1}{3}$$

$$\Rightarrow \tan A = -3$$

$$\tan C = \tan(\pi - (A + B)) = -\tan(A + B)$$

$$= \frac{\tan A + \tan B}{\tan A \tan B - 1} = \frac{-3 + \frac{1}{3}}{-1 - 1} = \frac{4}{3}$$

$$\therefore \sum \tan A = -3 + \frac{1}{3} + \frac{4}{3} = -\frac{4}{3} = \prod \tan A$$

$$\begin{aligned} \sum \tan A \tan B &= \left(-3 \times \frac{1}{3}\right) + \left(\frac{1}{3} \times \frac{4}{3}\right) + \left(\frac{4}{3} \times -3\right) \\ &= -1 + \frac{4}{9} - 4 = -\frac{41}{9} \end{aligned}$$

$$\sin^2(A + B) + \cos^2 C = 1$$

2. Ans. (B,C,D)

Sol. Let $A_1 = 3^r \cdot 5^s$

r, s are non negative integers

$$\Rightarrow \frac{1}{A_1} + \frac{1}{A_2} + \frac{1}{A_3} + \dots + \frac{1}{A_n} < \left(1 + \frac{1}{3} + \frac{1}{3^2} + \dots\right)$$

$$\left(1 + \frac{1}{5} + \frac{1}{5^2} + \dots\right) = \frac{1}{\left(1 - \frac{1}{5}\right)} = \frac{1}{\left(1 - \frac{1}{5}\right)}$$

$$= \frac{3}{2} \cdot \frac{5}{4} = \frac{15}{8}$$

3. Ans. (A,B,C)

Sol. $abcabc = a \times 10^5 + b \times 10^4 + c \times 10^3$
 $+ a \times 10^2 + b \times 10 + c$

$$= (100a + 10b + c) (1001)$$

$$= 7.11.13 (100a + 10b + c)$$

'abcabc' is divisible by 7, 11, 13, 77,

4. Ans. (B,C,D)

Sol. $kx + \lambda y = k + \lambda$

$$\lambda x + ky = 1$$

For system to be inconsistent

$$\frac{k}{\lambda} = \frac{\lambda}{k} \neq k + \lambda$$

$$\Rightarrow k^2 = \lambda^2 \text{ \& } k \neq k\lambda + \lambda^2$$

$$\Rightarrow k = \lambda \text{ or } k = -\lambda$$

Case I : If $k = \lambda$

$$\Rightarrow k \neq k^2 + k^2$$

$$\Rightarrow k \neq 0, \frac{1}{2}$$

Case II : If $k = -\lambda$

$$\Rightarrow k \neq -k^2 + k^2$$

$$\Rightarrow k \neq 0$$

$\therefore$ system is inconsistent

$$\text{If } k = \lambda \neq 0, \frac{1}{2}$$

$$\Rightarrow k + \lambda \neq 1$$

$\therefore$ (C) is incorrect & (A) is correct.

From Case II : $\frac{k}{\lambda} = -1$

$\therefore$ B is incorrect

If 2 points satisfy

$\Rightarrow$ Infinite solutions

$$\Rightarrow \frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

$$\Rightarrow \frac{k}{\lambda} = \frac{\lambda}{k} = k + \lambda$$

$$\Rightarrow k = \lambda = \frac{1}{2} \text{ is the only solution}$$

5. Ans. (B,C)

Sol. $C_1 \rightarrow C_1 + C_3, C_2 \rightarrow C_2 + C_3$

$$\Delta = \begin{vmatrix} a\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right) & b\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right) & 1 \\ b\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right) & c\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right) & 1 \\ c\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right) & a\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right) & 1 \end{vmatrix}$$

$$= \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right)^2 \begin{vmatrix} a & b & 1 \\ b & c & 1 \\ c & a & 1 \end{vmatrix}$$

$$= \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right)^2 [a^2 + b^2 + c^2 - ab - bc - ca] = 0$$

$$\Rightarrow a^2 + b^2 + c^2 - ab - bc - ca = 0$$

$$\text{or } (a - b)^2 + (b - c)^2 + (c - a)^2 = 0 \Rightarrow a = b = c$$

ALLEN**6. Ans. (A,B,C,D)****Sol.** Distance between the line is

$$2x + y - \frac{5}{2} = 0$$

$$2x + y - 1 = 0$$

$$\lambda = \frac{3/2}{\sqrt{5}} = \frac{3}{2\sqrt{5}}$$

$$\cos 120^\circ = \frac{a^2 + a^2 - \lambda^2}{2a^2}$$

$$\Rightarrow 3a^2 = \lambda^2 \Rightarrow a^2 = \frac{\lambda^2}{3} = \frac{3}{20}$$

area of hexagon is

$$6\left(\frac{\sqrt{3}}{4}a^2\right) = 6\left(\frac{\sqrt{3}}{4} \times \frac{3}{20}\right) = \frac{9\sqrt{3}}{40}$$

(B) Distance of centre from hexagon is

$$\frac{\lambda}{2} = \frac{3}{4\sqrt{5}}$$

(C) $2x + y - 1 = 0$

$$2x + y - \frac{5}{2} = 0$$

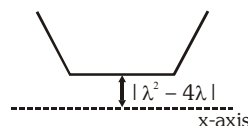
line equidistant from both is

$$\Rightarrow 2x + y - \left(\frac{5}{2} + 1\right) = 0$$

$$\Rightarrow 2x + y - \frac{7}{4} = 0$$

$$\Rightarrow 8x + 4y - 7 = 0$$

$$(D) \text{ Perimeter} = 6a = 6\sqrt{\frac{3}{20}} = \sqrt{\frac{36 \times 3}{20}} = \sqrt{\frac{27}{5}}$$

SECTION-II**1. Ans. 8.00****Sol.** The graph of the given equation is of type

or



$$\lambda \neq 0, 4$$

for infinite solution.

$$|\lambda^2 - 4\lambda| = |\lambda|$$

$$(\lambda^2 - 4\lambda)^2 - \lambda^2 = 0$$

$$(\lambda^2 - 5\lambda)(\lambda - 3\lambda) = 0$$

$$\lambda = 0, 3, 5 \text{ but } \lambda \neq 0$$

hence only possible value of λ are 3 & 5

hence answer is 8.

$$\text{if } \lambda = 0, 4$$

infinite solution

not possible.

2. Ans. 18.00**Sol.** $AB : x + y = 5$, $BC : x - 5y = 5$ & $AC : y = x - 1$ $\therefore$ Reflection of A in BC can be obtained as

$$\frac{x-3}{1} = \frac{y-2}{-5} = \frac{-2(3-10-5)}{26} \Rightarrow P\left(\frac{51}{13}, \frac{-34}{13}\right)$$

Similarly Q(1, 4) & R(6, 5)

 $\therefore$ Area of ΔPQR

$$= \frac{1}{2} \begin{vmatrix} \frac{51}{13} & -\frac{34}{13} & 1 \\ 1 & 4 & 1 \\ 6 & 5 & 1 \end{vmatrix} = 18 \text{ sq. units}$$

3. Ans. 128.00**Sol.** Now $y = x^7 + 64x^2$

$$= x^4(\underbrace{x^3 + 4x - 8}_{\text{zero}}) - 4x^5 + 8x^4 + 64x^2$$

$$= -4x^5 + 8x^4 + 64x^2$$

$$= -4x^2(\underbrace{x^3 + 4x - 8}_{\text{zero}}) + 8x^4 + 16x^3 + 32x^2$$

$$= 8x^4 + 16x^3 + 32x^2$$

$$= 8x(\underbrace{x^3 + 4x - 8}_{\text{zero}}) + 16x^3 + 64x$$

$$= 16(x^3 + 4x - 8) + 128 = 128$$

4. Ans. 72.00**Sol.** Draw parallelogram OBDC.

OE = ED & as

$$OC^2 = OD^2 + DC^2,$$

 ΔODC is aright angled $\Delta(6, 8, 10)$

$$\Rightarrow \text{Area of } \Delta ADC$$

$$= \frac{1}{2} \times 8 \times 12 = 48$$

$$\Delta EDC = \frac{1}{2} \times 3 \times 8 = 12$$

$$\text{Area } \Delta AEC = \text{Area } \Delta ADC - \text{Area } \Delta EDC$$

$$= 48 - 12 = 36$$

$$\text{Area } \Delta AEC = 36 \Rightarrow \text{Area } \Delta ABC = 72$$

5. Ans. 41.00

$$\text{Sol. } \alpha + \beta + \gamma + \delta = 20 \quad \dots\dots\dots (i)$$

$$\alpha\beta\gamma\delta = -1992$$

$$\text{If } \delta\gamma = 24$$

$$\Rightarrow \alpha\beta = \frac{-1992}{24} = -83 \quad \dots\dots\dots (ii)$$

$$\gamma\delta(\alpha + \beta) + \alpha\beta(\gamma + \delta) = -590$$

$$24(\alpha + \beta) - 83(\gamma + \delta) = -590$$

$$24 \times \{(\alpha + \beta) + (\gamma + \delta) = 20\}$$

$$-107(\gamma + \delta) = -1070 \Rightarrow \gamma + \delta = 10$$

$$\Rightarrow \alpha + \beta = 10$$

$$\text{so } (\alpha + \beta)(\gamma + \delta) + \alpha\beta + \gamma\delta = k$$

$$k = 10 \cdot 10 + 24 - 83 = 41$$

6. **Ans. 96.00**

Sol. Coordinates of any points lying on the line

$$y = \sqrt{3}x \text{ will be } \left(\frac{r}{2}, \frac{r\sqrt{3}}{2} \right)$$

If the given line intersects the curve

$$x^4 + ax^2y + bxy + cx + dy + 6 = 0, \text{ then}$$

$$\frac{r^4}{16} + a \frac{r^3\sqrt{3}}{8} + \frac{br^2\sqrt{3}}{4} + \frac{cr}{2} + \frac{dr\sqrt{3}}{2} + 6 = 0$$

$$r^4 + r^3 2a\sqrt{3} + r^2 4b\sqrt{3} + r 8(c + d\sqrt{3}) + 96 = 0$$

$$\therefore r_1 r_2 r_3 r_4 = 96$$

$$\text{hence } OA \cdot OB \cdot OC \cdot OD = 96$$

SECTION-III

1. **Ans. 4**

$$\text{Sol. } \frac{x}{a} + \frac{y}{b} = 1 \Rightarrow \frac{3}{a} + \frac{2}{b} = 1 \Rightarrow \frac{2}{b} = 1 - \frac{3}{a}$$

$$\Rightarrow b = \frac{2a}{a-3} = 2 \left(\frac{(a-3)+3}{(a-3)} \right) = 2 + \frac{6}{(a-3)}$$

$$a = 4, b = 8 \text{ \& } a = 2 \text{ but } b = -4$$

$$a = 5, b = 5 \quad a = 1 \text{ but } b = -1$$

$$a = 6, b = 4 \quad a = 0 \text{ but } b = 0$$

$$a = 9, b = 3$$

2. **Ans. 7**

$$\text{Sol. } \therefore \angle A = 100^\circ \Rightarrow \angle B = \angle C = 40^\circ$$

$$\text{Now, } (x^2 + y^2) = [(\sin A + \sin B + \sin C)^2 + (\cos A + \cos B + \cos C)^2]$$

$$= 1 + 1 + 1 + 2\sin A \sin B + 2\sin B \sin C + 2\sin C \sin A + 2\cos A \cos B + 2\cos B \cos C + 2\cos C \cos A$$

$$= 3 + 2\{(\cos A \cos B + \sin A \sin B) + (\cos B \cos C + \sin B \sin C) + (\cos A \cos C + \sin A \sin C)\}$$

$$= 3 + 2\{\cos 60^\circ + \cos 0^\circ + \cos(-60^\circ)\}$$

$$= 3 + 2\left\{\frac{1}{2} + 1 + \frac{1}{2}\right\} = 7$$

3. **Ans. 4**

$$\text{Sol. As } \cot\left(\frac{\pi}{3\sqrt{3}} \sin 2x\right) = \sqrt{3}, \text{ we get}$$

$$\frac{\pi}{3\sqrt{3}} \sin 2x = n\pi + \frac{\pi}{6}, n \in \mathbb{I}$$

$$\Rightarrow \sin 2x = 3\sqrt{3}n + \frac{\sqrt{3}}{2}, n \in \mathbb{I}$$

$$\Rightarrow \text{only possible value of } n \text{ is } 0 \text{ as}$$

$$\sin 2x \in [-1, 1]$$

$$\text{Hence, } \sin 2x = \frac{\sqrt{3}}{2}$$

$$\text{Hence, } 2x = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{7\pi}{3}, \frac{8\pi}{3}$$

$$\text{Which given } x = \frac{\pi}{6}, \frac{\pi}{3}, \frac{7\pi}{6}, \frac{4\pi}{3} \text{ (4 solutions)}$$

4. **Ans. 9**

Sol. Using m-n cot theorem in $\triangle ABC$

$$a \cot \theta = \frac{a}{2} \cot C - \frac{a}{2} \cot B$$

$$\Rightarrow \cot \theta = \frac{\cot C - \cot B}{2} = \frac{\sin(B-C)}{2 \sin B \sin C} = \frac{\sin A \sin(B-C)}{2 \sin A \sin B \sin C}$$

$$= \frac{\sin^2 B - \sin^2 C}{2 \sin^2 A \frac{\sin B \sin C}{\sin A}} = \frac{b^2 - c^2}{4 \left(\frac{1}{2} a^2 \frac{\sin B \sin C}{\sin A} \right)} = \frac{b^2 - c^2}{4\Delta}$$

$$\Rightarrow \cot \theta = \frac{25-16}{4 \cdot 8} \Rightarrow 32 \cot \theta = 9$$

5. **Ans. 4**

Sol. $P(x) = ax^3 + bx^2 + cx + d$

$$P\left(\frac{1}{2}\right) + P\left(-\frac{1}{2}\right) = \left(\frac{b}{4} + d\right) 2 = \left(\frac{b}{2} + 2d\right)$$

$$\frac{P\left(\frac{1}{2}\right) + P\left(-\frac{1}{2}\right)}{P(0)} = \frac{\left(\frac{b}{2} + 2d\right)}{d} = \frac{b}{2d} + 2 = 100$$

$$\frac{b}{d} = 196$$

$$\frac{(\alpha + \beta + \gamma)}{\alpha\beta\gamma \times 49} = \frac{-b/a}{\left(-\frac{d}{a}\right) 49} = \frac{b}{d \times 49} = \frac{196}{49} = 4$$

6. **Ans. 2**

Sol. $\theta_1 - \theta = \theta - \theta_2$

$$2\theta = \theta_1 + \theta_2$$

$$\tan 2\theta = \frac{\tan \theta_1 + \tan \theta_2}{1 - \tan \theta_1 \tan \theta_2}$$

$$= \frac{7+1}{1-7 \cdot 1} = -4/3$$

$$\frac{2 \tan \theta}{1 - \tan^2 \theta} = \frac{-4}{3}$$

$$\Rightarrow 2 \tan^2 \theta - 3 \tan \theta - 2 = 0$$

$$\Rightarrow (2 \tan \theta + 1)(\tan \theta - 2) = 0$$

$$\Rightarrow \tan \theta = 2 \text{ or } -1/2$$

$$\therefore \text{Slope of longer diagonal is } 2.$$

